ExpenseAI

Complete Features & Components Documentation

Version 2.0 | AI-Powered Expense Management

📊 Application Overview

ExpenseAI is a comprehensive expense management application built with React, featuring AI-powered insights, receipt scanning, multi-currency support, and advanced analytics. The application uses a backend server for data persistence and authentication.

🚀 Latest Features (Recent Updates)

Advanced OCR Receipt Scanning: Enhanced image preprocessing with adaptive thresholding, noise reduction, and AI-based text correction for worst-condition receipts.

Predictive Reading: Context-aware character correction and receipt-specific term fixing for improved accuracy.

Multi-Currency Support: Real-time currency conversion with exchange rate updates.

AI Chatbot: Local AI assistant with voice input, financial insights, and pattern matching.

🧩 Core Components

App.tsx Main

Root component managing application state, routing, and authentication flow.

- ✓ User authentication and session management
- ✓ Multi-view navigation (Dashboard, Analytics, Settings, etc.)
- ✓ Dark/Light theme toggle with persistence
- ✓ Currency conversion and display
- ✓ Search functionality for expenses
- ✓ Expense and income data management

Sidebar.tsx

Navigation sidebar with collapsible functionality and user profile display.

- ✓ Collapsible sidebar with width adjustment
- ✓ User email display
- ✓ Navigation to all main views
- ✓ Logout functionality
- ✓ Responsive design

Login.tsx Auth

Authentication component with login, signup, forgot password, and reset password views.

- ✓ Login with email and password
- ✓ Registration with email validation
- ✓ Forgot password with OTP generation
- ✓ Password reset functionality
- ✓ Animated background with rotating images
- ✓ Glassmorphism UI design
- ✓ Session management with token storage

💰 Expense Management

ExpenseForm.tsx OCR

Form for adding new expenses with AI-powered category suggestions and receipt scanning integration.

- ✓ Receipt scanning with OCR integration
- ✓ AI-powered category suggestions based on description
- ✓ Tax information tracking
- ✓ Date picker with today's date default
- ✓ Notes field for additional details
- ✓ Real-time validation

- ✓ Scan confidence indicators
- ✓ Auto-fill from receipt scan results

ExpenseList.tsx

List view for all expenses with filtering, sorting, and editing capabilities.

- ✓ Display all expenses in a list format
- ✓ Edit and delete functionality
- ✓ Category-based filtering
- ✓ Date-based sorting
- ✓ Currency display with symbols
- ✓ Responsive card layout

ExpenseTimelineGraph.tsx Analytics

Advanced timeline visualization for expense tracking with multiple view options.

- ✓ Multiple timeline views (Today, This Week, This Month, This Year, 24h, 7d, 30d, 90d, 1y, All Time)
- ✓ Search functionality within timeline
- ✓ Statistics display (total, average, min, max)
- ✓ Trend indicators (up/down arrows)
- ✓ Category filtering
- ✓ Interactive expense cards

- ✓ Dark mode support

💵 Income Management

IncomeForm.tsx

Form for adding new income entries.

- ✓ Description input
- ✓ Amount field with currency symbol
- ✓ Category selection
- ✓ Date picker
- ✓ Notes field
- ✓ Real-time validation

IncomeList.tsx

List view for all income entries with edit and delete capabilities.

- ✓ Display all income entries
- ✓ Edit and delete functionality
- ✓ Category-based display
- ✓ Date sorting

- ✓ Currency formatting

IncomeOverview.tsx

Overview component displaying income statistics and breakdown.

- ✓ Total income display
- ✓ Category-wise breakdown
- ✓ Monthly income trends
- ✓ Visual indicators

🤖 AI Features

AIChatbot.tsx AI

Advanced AI chatbot with local processing, voice input, and financial insights.

- ✓ Local AI with pattern matching (no external API required)
- ✓ Voice input using Web Speech API
- ✓ Text-to-speech output
- ✓ Financial context awareness
- ✓ Conversation history tracking
- ✓ Quick action suggestions
- ✓ Helpful/unhelpful feedback system

- ✓ Copy message functionality
- ✓ Streaming response simulation
- ✓ Expense and income analysis
- ✓ Budget recommendations
- ✓ Savings advice

aiEngine.ts AI

Core AI engine for category suggestions and insights generation.

- ✓ Keyword-based category classification (100+ keywords per category)
- ✓ Confidence scoring for suggestions
- ✓ Pattern matching for expense descriptions
- ✓ Insight generation (warnings, tips, info)
- ✓ Category totals calculation
- ✓ Spending pattern analysis
- ✓ Indian-specific keywords (biryani, dosa, sambar, etc.)
- ✓ Delivery app recognition (Zomato, Swiggy, Blinkit, etc.)

ReceiptScanPanel.tsx OCR Enhanced

Advanced receipt scanning with enhanced OCR and AI text correction.

- ✓ **Advanced Image Preprocessing:**
  - ✓ High-resolution scaling (up to 3000px)

- ✓ Multi-step processing: grayscale, contrast, brightness, saturation
- ✓ Adaptive thresholding simulation
- ✓ Pixel-level processing for text contrast
- ✓ Sharpening and final binarization

✓ **AI-Based Text Correction:**

- ✓ Context-aware character correction (0/O, 1/I/l, 5/S, 8/B)
- ✓ Receipt-specific term fixing (TOTAL, AMOUNT, GST, CGST, etc.)
- ✓ Currency symbol normalization (Rs., INR → ₹)
- ✓ Decimal point correction
- ✓ Extra space removal

✓ **Enhanced Tesseract Configuration:**

- ✓ LSTM_ONLY engine for better accuracy
- ✓ Advanced page segmentation
- ✓ Dictionary and frequency dictionary enabled
- ✓ Better text line detection
- ✓ Adaptive thresholding
- ✓ Improved character classification
- ✓ Better noise handling
- ✓ Enhanced word segmentation

✓ Live camera capture with real-time preview

✓ Gallery image upload

- ✓ Review modal with editable results
- ✓ Confidence level indicators (high/medium/low)
- ✓ Warning messages for uncertain data
- ✓ Tax information extraction (GST, CGST, SGST, IGST, VAT)
- ✓ Item count detection
- ✓ Both file upload and webcam support

receiptParser.ts OCR

Receipt text parsing with merchant detection and line item extraction.

- ✓ Merchant name detection
- ✓ Total amount extraction with scoring
- ✓ Line item extraction
- ✓ Tax information extraction
- ✓ Confidence level calculation
- ✓ Warning generation for uncertain data
- ✓ Indian receipt format support
- ✓ Multiple currency symbol support

📈 Analytics & Visualization

CategoryPieChart.tsx

Pie chart visualization for expense categories.

- ✓ Interactive pie chart
- ✓ Category-wise spending breakdown
- ✓ Percentage calculation
- ✓ Color-coded categories
- ✓ Hover effects
- ✓ Legend display

SpendingChart.tsx

Spending trend visualization.

- ✓ Line chart for spending trends
- ✓ Time-based data visualization
- ✓ Responsive design

StatsOverview.tsx

Overview statistics dashboard.

- ✓ Total expenses display
- ✓ Total income display
- ✓ Net savings calculation

- ✓ Monthly averages
- ✓ Quick stats cards

InsightsPanel.tsx

AI-generated insights panel.

- ✓ Warning insights
- ✓ Tips and recommendations
- ✓ Informational messages
- ✓ Icon-based categorization

🎯 Budget & Goals

BudgetProgressTracker.tsx New

Budget tracking with progress indicators and projections.

- ✓ Monthly budget tracking
- ✓ Progress bar visualization
- ✓ Projected monthly spending calculation
- ✓ Daily average calculation
- ✓ Days remaining countdown
- ✓ Budget alerts at 90% usage

- ✓ Remaining budget per day calculation

SavingsGoalProgress.tsx New

Savings goals tracking with progress visualization.

- ✓ Multiple savings goals (Emergency Fund, Vacation, New Phone, Investment)
- ✓ Progress bars for each goal
- ✓ Achievement indicators (star icons)
- ✓ Total savings display
- ✓ Goal target amounts
- ✓ Percentage completion calculation
- ✓ Visual goal icons

MonthlyComparison.tsx New

Month-over-month expense comparison.

- ✓ Current month vs previous month comparison
- ✓ Percentage change calculation
- ✓ Trend indicators (up/down)
- ✓ Category-wise comparison

⚡ Quick Actions

QuickActions.tsx New

Quick action buttons for common tasks.

- ✓ Add expense shortcut
- ✓ Add income shortcut
- ✓ View analytics shortcut
- ✓ Quick access to common features

QuickSummary.tsx New

Quick summary dashboard with key metrics.

- ✓ Total expenses summary
- ✓ Total income summary
- ✓ Net savings display
- ✓ Quick stats cards
- ✓ At-a-glance financial overview

⚙️ Settings & Account

AccountSettings.tsx

Comprehensive account settings panel.

- ✓ Profile picture upload and management
- ✓ Account information display (email, creation date)
- ✓ Statistics overview (total expenses, income, counts)
- ✓ Currency selection with real-time conversion
- ✓ Exchange rate refresh functionality
- ✓ Dark/Light theme toggle
- ✓ Password change functionality
- ✓ Groq API key configuration for AI features
- ✓ Data export (expenses and income)
- ✓ Account deletion with confirmation
- ✓ Logout functionality

UserProfile.tsx

User profile display component.

- ✓ Profile picture display
- ✓ User email display
- ✓ Account creation date
- ✓ Edit profile functionality

🎨 UI Components

button.tsx

Reusable button component with variants.

- ✓ Multiple button variants (default, destructive, outline, ghost)
- ✓ Size options (default, sm, lg, icon)
- ✓ Loading state support
- ✓ Disabled state

card.tsx

Card component for content grouping.

- ✓ Card header, title, and content
- ✓ Responsive design
- ✓ Dark mode support

input.tsx

Input field component with validation.

- ✓ Text input with placeholder
- ✓ Disabled state
- ✓ Dark mode support

label.tsx

Label component for form fields.

- ✓ Form label with htmlFor attribute
- ✓ Dark mode support

progress.tsx

Progress bar component.

- ✓ Configurable progress value
- ✓ Customizable height
- ✓ Dark mode support

🔧 Utilities

helpers.ts

Helper functions for common operations.

- ✓ Currency formatting with symbols
- ✓ Date formatting
- ✓ Currency conversion with exchange rates
- ✓ Exchange rate refresh functionality
- ✓ Preferred currency management

- ✓ ID generation
- ✓ Today's date string
- ✓ Expense sorting by date

currencyConverter.ts

Currency conversion utilities.

- ✓ Exchange rate fetching from API
- ✓ Currency conversion functions
- ✓ Rate caching
- ✓ Error handling

types/index.ts

TypeScript type definitions.

- ✓ Expense interface
- ✓ Income interface
- ✓ AiSuggestion interface
- ✓ CategoryTotal interface
- ✓ Insight interface

🖥️ Backend (Python)

server.py

Python backend server for data persistence and authentication.

- ✓ SQLite database support
- ✓ PostgreSQL support (optional)
- ✓ User authentication (register, login, logout)
- ✓ Password reset with OTP
- ✓ Expense CRUD operations
- ✓ Income CRUD operations
- ✓ Session management
- ✓ CORS support
- ✓ Email configuration for password reset
- ✓ SMTP configuration management
- ✓ API endpoint routing
- ✓ Error handling

api.ts (Frontend API Client)

Frontend API client for backend communication.

- ✓ HTTP request wrapper
- ✓ Token-based authentication
- ✓ Error handling
- ✓ User authentication functions

- ✓ Expense management functions
- ✓ Income management functions
- ✓ Password reset functions
- ✓ Session management

📋 Features Summary Table

Feature	Status	Description
Receipt OCR Scanning	Enhanced	Advanced OCR with AI text correction and preprocessing
AI Chatbot	Active	Local AI with voice input and financial insights
Multi-Currency	Active	Real-time currency conversion with exchange rates
Budget Tracking	Active	Monthly budget tracking with projections
Savings Goals	Active	Multiple savings goals with progress tracking
Expense Timeline	Active	Advanced timeline with multiple view options
Category Suggestions	Active	AI-powered category suggestions based on description

Feature	Status	Description
Dark Mode	Active	Dark/Light theme toggle with persistence
Authentication	Active	Login, signup, password reset with OTP
Data Export	Active	Export expenses and income data

🛠️ Technology Stack

Category	Technologies
Frontend	React, TypeScript, Vite, TailwindCSS
Backend	Python, SQLite/PostgreSQL, HTTP Server
OCR	Tesseract.js with enhanced preprocessing
AI	Local pattern matching, Groq API (optional)
Icons	Lucide React
Charts	Custom SVG-based visualizations
Build	Vite, TypeScript Compiler
Desktop	Electron (for desktop app packaging)

ExpenseAI - AI-Powered Expense Management

Generated on June 11, 2026 | Version 2.0